Model Program Book

SHORTTERM INTERNSHIP

(Virtual)

Designed & Developed by

ANDHRA PRADESH

STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF GOVERNMENT OF ANDHRA PRADESH)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

(Virtual)

Name of the Student: Thota Venkata Rao

Name of the College: Lingayas Institute of Management and Technology

Registration Number: 22NA1A6103

Period of Internship: From: 20/05/25 To: 06/07/25

Name & Address of the Intern Organization

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YEAR
2025

An Internship Report on

Java Full Stack

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Prof. J. Bindu Kumar

Department of

Artificial Intelligence & Machine learning

Lingayas Institute of Management & Technology

Submitted by:

Thota Venkata Rao

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Student's Declaration

I am Thota Venkata Rao a student of Bachelor of Technology Program, Reg. No: 22NA1A6103 of the Department of Artificial Intelligence & Machine learning College do hereby declare that I have completed the mandatory internship from 20/05/2025 to 06/07/2025 in Sandspace Technologies Private Limited under the Faculty Guideship of Prof. J. Bindu Kumar, Department of Artificial Intelligence & Machine learning, Lingayas Institute of Management and Technology.

(Signature and Date)

Official Certification

This is to certify Thota Venkata Rao that Reg. No.22NA1A6103 has completed his/her Internship in Sandspace Technologies Private Limited on Full Stack Java under my supervision as a part of partial fulfillment of the requirement for the Department of Artificial Intelligence & Machine learning, Lingayas Institute of Management and Technologies.

(Signatory with Date and Seal)

Endorsements

Faculty Guide

Head of the Department

Principal

Certificate from Intern Organization

TEGA ISO 9001:2015 nasscom

CERTIFICATE

of COMPLETION

This is to certify that Thota Venkatrao

has successfully completed an internship in the

Java Full stack at Sandspace Technologies Pvt Ltd

from May 2025 to July 2025.

We acknowledge the valuable contribution made during the internship and extend our best wishes for future achievements.

HR MANAGER

Acknowledgements

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Thota Venkata Rao
(22NA1A6103)

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CHAPTER 1: EXECUTIVE SUMMARY

The report epitomizes about internship performed with the time span of two months in Sandspace Technologies Private Limited by Thota Venkata Rao. It is one of the companies approved for providing meaningful industrial exposure as per the guidelines of the All-India Council for Technical Education (AICTE) and Andhra Pradesh State Council of Higher Education (APSCHE). An intern needs to do his/her internship in AICTE or APSCHE recognized companies only.

1.1 This internship has paved a way to plan for a few learning objectives that include,

- Get exposed to professional tasks and job roles in a reputed software organization and gain hands-on experience.
- Develop both technical and communication skills in the chosen field of work.
- Achieve working experience both with the team and individually.
- Enhance technical, analytical, and soft skills through active involvement in live projects.
- Cultivate problem-solving and critical thinking abilities to address challenges encountered during the internship.

1.2 The following are the learning outcomes out of the learning objects framed:

- The internship helped in opting for a particular track, task & provided real-time exposure to software development tasks and enhanced practical knowledge.
- This internship aided in effectively communicating with peers for technical decisions.
- This internship paved a path to experience working with team & individually.
- Strengthened technical skills in programming and software tools, while also improving teamwork, time management, and interpersonal skills.
- This internship also aided in developing networking contacts in the organization.

Overall, the internship at Sandspace Technologies has been a highly enriching experience that contributed to both technical and personal growth. It provided a platform to explore advanced software development practices, real-time project handling, and organizational culture. This internship not only improved problem-solving and teamwork skills but also gave exposure to cutting-edge, real-time projects, which will be valuable in selecting and executing future academic or professional projects.

CHAPTER 2: OVERVIEW OF THE ORGANIZATION

Sandspace Technologies Private Limited is a distinguished and successful software company based in Vijayawada, Andhra Pradesh, with a legacy of over 16 years in the software industry. The company has consistently demonstrated leadership in developing impactful, real-time projects that contribute to national development and technological advancement. With an annual turnover of ₹2.1 crore, Sandspace has established strong financial stability and sustained profitability, reflecting its commitment to innovation, quality, and sustainable growth.

2.1 Training / Internship Areas:

Sandspace provides opportunities for learning and professional development through its comprehensive internship and training programs in various domains, including:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Web Development
- Cloud Computing
- Software Development and Testing
- Data Analytics
- Full Stack Java
- Mobile App Development

2.2 Training / Internship Methods:

Sandspace emphasizes hands-on learning by directly involving interns in real-time projects. The internship program follows a practical approach where students gain exposure to live projects under the mentorship of experienced professionals. Along with technical guidance, the company also provides access to tools, learning platforms, and industry-level practices.

<u>Faculty / Mentors:</u>

Interns are assigned dedicated mentors from the professional team who provide:

- Technical guidance and project specifications
- Regular progress reviews and feedback
- Code reviews and constructive suggestions
- Career development insights and skill-building support

2.3 Placement / Career Opportunities:

Sandspace uses its internship program as a talent pipeline, encouraging high-performing interns to explore professional roles within the organization. Its strong industry presence and focus on impactful projects help students gain practical skills and improve employability in the competitive job market.

Location:

Sandspace Technologies Private Limited is located in Vijayawada, Andhra Pradesh, India.

Vision:

The vision of Sandspace Technologies is to establish itself as a global leader in real-time software solutions, empowering national progress and transforming productivity in the technological landscape.

Mission:

The mission is centered on designing, developing, and delivering advanced software solutions that address critical national needs. The company is committed to technical excellence, innovation, and impactful contributions to both clients and society.

2.4 Core Values:

- Innovation
- Excellence
- Integrity
- National Impact
- Productivity Enhancement
- Collaboration & Teamwork
- Customer-Centric Approach
- Continuous Learning

2.5 Services Provided by the Organization:

- Real-time Software Development
- AI and ML-based Applications
- Web and Cloud Solutions
- Data Analytics and Business Intelligence
- Customized Project Development for National and Industrial Needs

2.6 Strategy:

- **Quality Delivery:** Focus on producing reliable and high-quality software solutions.
- **Innovation:** Encouraging creativity and fresh ideas to solve real-world challenges.
- **Hands-on Experience:** Emphasizing practical project involvement for interns and employees.
- **Collaboration:** Building strong teamwork and client partnerships.
- **National Development:** Aligning projects with the vision of contributing to India's digital growth.

2.7 Structure of the Organization:

Sandspace follows a hybrid organizational model, which includes both functional departments and project-based teams. The leadership structure consists of:

- Chief Executive Officer (CEO)
- Chief Technology Officer (CTO)
- Chief Operating Officer (COO)
- Departments: Software Development, Quality Assurance, Project Management, Business Development, HR & Administration, and IT Infrastructure Support.

2.8 Roles and Responsibilities:

- Developing and delivering innovative software solutions
- Conducting real-time project work
- Mentoring and guiding interns
- Providing technical and analytical support to clients
- Fostering teamwork and knowledge sharing

2.9 Future Plans of the Organization:

Sandspace Technologies plans to expand its research and development division, explore new service domains, and enter emerging global markets. The company aims to invest in cutting-edge infrastructure, strengthen employee training, and continue developing impactful projects in AI, ML, and other emerging technologies. With its strong vision, Sandspace aspires to maintain its leadership in the software industry while contributing to both national and global technological growth.

CHAPTER 3: INTERNSHIP PART

3.1 Description of Activities/Responsibilities

This chapter provides a detailed account of my internship journey at Sandspace Technologies Private Limited, where I served as a Full Stack Java Intern for a duration of two months. During this internship, I was assigned a comprehensive project titled "E-Platform for Library Resources", an advanced and digitized library management system designed to streamline library services and enhance accessibility for students, faculty, and administrators.

The internship aimed to bridge the gap between theoretical learning and practical implementation by exposing me to real-world development environments, project lifecycle management, and modern software engineering practices. Under the guidance of my mentor, I gained valuable insights into full stack development, debugging, testing, and deployment of web-based applications.

3.2 Working Conditions

The internship was conducted in a remote setup, with access to all the required digital tools and platforms. Regular mentorship and code review sessions were organized via online meetings, ensuring continuous learning and guidance throughout the project. Collaborative discussions and technical reviews helped me resolve challenges and enhance the functionality of the system.

Work Environment Characteristics:

- Remote working mode with virtual collaboration tools.
- Weekly mentor-led feedback sessions and sprint reviews.
- Code deployment discussions and debugging sessions.
- Progress tracking through documentation and Git-based version control.

3.3 Weekly Work Schedule

Throughout the internship, I dedicated approximately 49 hours per week, ensuring a balanced approach between learning new concepts and executing project deliverables. Each week was structured around milestone-based goals, such as frontend design, backend development, API testing, and system integration. This organized schedule facilitated steady progress and incremental improvements to the platform's features.

3.4 Technical Environment

The development of the E-Platform for Library Resources required a well-defined technical environment comprising both hardware and software components.

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 500 GB
- Speed: 2.4 GHz or higher

Software Requirements

- **Operating System:** Windows 10/11
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Java, Spring Boot, Maven
- **Database:** MySQL, Spring Data JPA
- **Testing Tools:** Postman (API testing), MySQL Workbench
- **IDE:** IntelliJ IDEA / Eclipse / Vs code

3.5 Activity Title

"E-Platform to access library resources".

3.6 Activity Overview

The E-Platform for Library Resources is a web-based, role-driven application developed to simplify and automate the day-to-day operations of a college library.

The system integrates multiple layers — frontend (presentation layer), backend (application logic), and database (data persistence) — to provide a seamless and efficient experience for all users.

The application provides three main roles:

- Admin: Manages library books, user records, borrowing requests, and fine approvals.
- Faculty: Accesses and borrows books using institutional email OTP authentication.
- Student: Browses, borrows, and returns books while managing fines for late returns.

This system bridges traditional library management with digital technology, providing real-time access to resources and reducing manual overhead.

3.7 Activity Objectives

The primary objectives of the project were to:

- Develop a responsive and scalable web application for library resource management.
- Implement role-based authentication and authorization for Admin, Faculty, and Student users.
- Integrate frontend and backend seamlessly through RESTful APIs.
- Design an optimized MySQL database schema for managing books, users, and borrow records.
- Validate and test APIs using Postman for consistency and accuracy.
- Gain hands-on exposure to the end-to-end software development life cycle (SDLC), including requirement analysis, development, testing, and deployment.

3.8 Activity Approach & Detailed Tasks

The entire development process followed an Agile methodology, focusing on modular and iterative implementation. Each component of the application was developed, tested, and integrated systematically.

1. Frontend Development (Client-Side)

The frontend was developed using HTML, CSS, and JavaScript, ensuring a clean and intuitive user experience.

Tasks Performed:

- Designed interactive and visually appealing pages, including login, dashboard, and catalog screens.
- Implemented JavaScript-based form validation, OTP input validation, and dynamic data rendering.
- Developed a responsive layout adaptable across desktop and mobile devices using CSS Flexbox and Grid.
- Created role-based navigation flows for Admin, Faculty, and Student users.

2. Backend Development (Server-Side)

The backend architecture was implemented using Java Spring Boot, adhering to modular design principles.

Key Implementations:

- Developed RESTful APIs to handle user authentication, book management, and borrow requests.
- Utilized Maven for dependency management and build automation.

- Adopted the Controller–Service–Repository layered architecture for maintainability and scalability.
- Integrated Spring Security for secure login and role-based access control.
- Configured JavaMailSender for sending OTPs to institutional email addresses.

3. Database Integration (Persistence Layer)

The system's data layer was managed using MySQL integrated with Spring Data JPA.

Database Design:

- Users Table: Stores admin, faculty, and student login credentials.
- Books Table: Contains book ID, title, author, branch, rack number, year, and availability status.
- Borrow Records Table: Maintains information about borrowed books, issue date, due date, and fine amount.
- OTP Table: Temporarily stores OTPs for verification.

Data Integrity:

- Implemented relational constraints such as Primary Key, Foreign Key, and NOT NULL.
- Ensured referential integrity between books and borrow records.
- Enabled automatic fine calculation for overdue books.

4. Testing and Debugging

Testing was performed to validate the functionality, performance, and reliability of each module.

Testing Activities:

- API Testing: Conducted using Postman for endpoints related to login, book management, and borrowing.
- Database Validation: Ensured proper data insertion, updates, and retrieval from MySQL tables.
- UI Testing: Verified frontend responsiveness, OTP verification flow, and dashboard interactivity.
- Debugging: Resolved issues in API responses, SQL queries, and component integration.

5. Features Implemented

Admin Module

- Add, delete, and update books.
- View and approve borrow requests.
- Track issued and returned books.
- Manage fine collection and view borrowing history.

Student Module

- View available books categorized by semester or branch.

- Borrow and return books.
- Pay fines through Razorpay integration.

Faculty Module

- Login through institutional email OTP.
- Borrow and return books without fines.
- View personalized borrowing history.

3.9 Implementation & Output

The E-Platform for Library Resources was developed through the integration of multiple modules working in unison—frontend, backend, and database layers. Each module was tested individually and then combined to form a cohesive system. The implementation outcomes are presented below, along with screenshots of major functional interfaces of the application.

3.9.1 Home Page – System Entry Interface

Figure 3.1: Homepage of E-Platform for Library Resources

The homepage acts as the gateway to the system. It displays navigation options for the three user categories—Admin, Faculty, and Student—and provides access to their respective login modules.

Process Description:

- Built using HTML and CSS with responsive design principles.
- Uses JavaScript to handle navigation routing.
- The background image and overlay text were designed to simulate a modern digital library experience.
- Clicking on "Admin," "Faculty," or "Student" dynamically redirects the user to their respective authentication modules.

This page also contains institutional information such as contact details, working hours, and copyright notice at the footer, providing an official appearance suitable for academic integration.

3.9.2 Admin Login Module

Figure 3.2: Admin Login Interface

The Admin Login page allows authorized personnel to access administrative functions such as managing books and users.

Technical Flow:

1. Admin enters credentials (username and password).
2. The frontend sends a POST request to /api/admin/login.
3. Backend (Spring Boot) validates credentials using the AdminRepository class.
4. On success, the session is initiated and redirects to the admin dashboard.
5. Invalid login attempts are handled with backend error messages displayed on the UI.

Backend Integration:

- Controller: AdminController.java
- Service: AdminService.java
- Repository: AdminRepository.java
- Database Table: admin_login (columns: username, password, role)

This secured login ensures that only authorized admins can access book management operations.

3.9.3 Faculty Login (OTP Authentication)

Figure 3.3: Faculty OTP-Based Login Interface

The faculty login implements email-based OTP authentication, enhancing security and eliminating password storage.

Process Flow:

1. Faculty enters their institutional email (e.g., name.lingayas@limat.edu.in).
2. System validates the domain and generates a 6-digit OTP.
3. The OTP is sent via Spring Boot Mail Service using JavaMailSender.
4. The OTP is stored temporarily in the MySQL table otp_verification with a timestamp.
5. Upon submission, the backend verifies OTP validity and expiry (5-minute limit).
6. Successful login redirects the user to the Faculty Dashboard.

Outcome:

This module ensures secure, institution-verified login without storing permanent credentials, aligning with data protection standards.

3.9.4 Student Login (OTP Authentication)

Figure 3.4: Student OTP Login Interface

The student login system functions similarly to faculty authentication but with domain-level validation for student IDs (22na1aXXXX@limat.edu.in).

Steps Involved:

1. Student provides institutional email.
2. OTP generated and sent to the registered email.
3. OTP verification handled through backend endpoint /api/student/verifyOTP.
4. On successful validation, student is redirected to the main dashboard.

Technical Integration:

- Java backend validates OTP using a secure random generator.
- OTP storage handled with automatic expiration and cleanup scheduler.
- MySQL maintains temporary session logs for security tracking.

This process ensures each student's identity is verified in real time, enhancing user accountability.

3.9.5 Main Dashboard – Role-Based Resource Access

Figure 3.5: Main Dashboard – Resource Categories

After authentication, users are redirected to the main dashboard. The dashboard displays resource categories like B.Tech Education, M.Tech/MBA Education, Career Development, Research, GitHub & Projects, Digital Resources, and Newspaper Archives.

Functional Workflow:

- The system fetches resource metadata from backend APIs (/api/resources).
- Each card represents a functional submodule.
- Clicking a category redirects to the corresponding resource page (e.g., Book Catalog, Career Hub).
- Logout button clears the session and redirects to the homepage.

Technologies Used:

- JavaScript for DOM event handling.
- Conditional rendering based on user role.
- Spring Boot session management for logout functionality.

3.9.6 Responsive Dashboard Design

Figure 3.6: Mobile-Optimized Dashboard View

This figure shows the responsive version of the dashboard optimized for mobile and tablet users.

Technical Features:

- CSS Flexbox and Media Queries used for automatic resizing.
- Dynamic grid layout to stack elements vertically on smaller screens.
- Optimized image compression for faster loading.

This ensures uniform usability and visual consistency across all devices, improving user accessibility.

3.9.7 B.Tech Catalog – Available Books

Figure 3.7: Book Catalog Page for B.Tech Students

The Book Catalog module allows students and faculty to explore available books categorized by program, year, and branch.

Process Flow:

1. The page dynamically fetches data from the books table using /api/books endpoint.
2. Dropdown filters enable users to refine search results.
3. “View” button opens book description and image in a modal.
4. “Borrow” button triggers a borrowing request that is stored in the borrow_records table.

Database Tables Used:

- books: Contains book metadata (title, author, rack, year).
- borrow_records: Tracks issue status and due date.

This module provides real-time visibility into library resources and book availability.

3.9.8 Borrowed Books and Fine Management

Figure 3.8: Borrowed Books with Fine Calculation

The borrowed books interface lists all active borrowings with details like issue date, return date, and fine amount.

Fine Management Logic:

- For students: ₹1 per day after 15 days of borrowing.
- For faculty: No fine applied.
- Fines are calculated automatically by backend scheduler (@Scheduled in Spring Boot).

Functional Workflow:

1. Data fetched from borrow_records.
2. Fines computed dynamically based on current date.
3. "Pay ₹1" or "Fine Paid" buttons dynamically update transaction status after payment.

This ensures transparency and accountability in the borrowing-return cycle.

3.9.9 Fine Payment Interface (Razorpay Integration)

Figure 3.9: Library Fine Payment Gateway

To simplify payment handling, the system integrates Razorpay for online fine payments.

Workflow:

1. Student clicks on "Pay ₹1."
2. Backend generates a payment order through Razorpay API.
3. Razorpay modal opens allowing UPI, card, or wallet transactions.
4. Payment confirmation received through webhook and updated in payment_records table.
5. Fine status automatically changes to "Paid."

Outcome:

This feature digitizes fine collection and ensures secure transactions through verified channels, eliminating manual cash handling.

3.9.10 Career Development Module

Figure 3.10: Career Hub Interface

The Career Development module extends the application beyond library management by integrating career guidance and job listings for students.

Functionalities:

- Resume Builder for students to create or update their profiles.
- Tabs for Aptitude, Coding, Interview, and Job Opportunities.
- Integrated REST APIs fetch real-time job openings.
- Download and export options for bookmarked opportunities.

Outcome:

This value-added component transforms the E-Platform into a comprehensive student resource hub, promoting both academic and career growth.

3.9.11 Resume Builder Module

Figure 3.11: Resume Builder Interface

The Resume Builder module enables students to create, preview, and download professional one-page resumes directly from the LMS.

Process Description:

1. Users enter details such as name, contact information, education, skills, and project links.
2. The frontend dynamically renders a real-time preview using JavaScript and browser-based storage.
3. Each resume element (header, experience, education, skills) is validated locally before proceeding to the next step.
4. The completed resume can be exported as PDF using the "Print/Save PDF" feature.

Technical Implementation:

- HTML and JS-based form builder with step-wise navigation.
- Client-side data storage for privacy (no backend storage).
- PDF export handled through the browser's print API.

Outcome:

Students can instantly generate ATS-optimized resumes, improving readiness for placements and internships while integrating seamlessly with the *Career Hub* module.

3.9.12 Research Repository

Figure 3.12: Research Hub — Repository Interface

The Research Repository acts as a digital archive where users can upload, view, and download research papers or technical articles.

Process Flow:

1. Data is fetched from the research_papers table through a Spring Boot REST API.
2. Each record contains the title, author, publication year, and access link.
3. The "View" button opens an in-app reader, while "Download" fetches the PDF from the

storage path.

4. The search bar allows keyword-based filtering for research papers.

Technical Details:

- Backend controller: ResearchController.java
- File management: handled via FileStorageService.
- Database: research_papers (id, title, author, year, filePath).

Outcome:

This module transforms the LMS into a knowledge-sharing platform, encouraging academic research, publication, and collaboration.

3.9.13 GitHub & Projects Hub

Figure 3.13: GitHub and Project Integration Module

This module provides students with an interface to share and explore academic projects hosted on GitHub.

Functional Workflow:

1. Users can link their GitHub repositories to the LMS for public display.
2. Projects can be filtered by year, semester, or technology stack (e.g., “4-1,” “3-2,” etc.).
3. Admins can verify and approve projects before publication.
4. The search and refresh functionalities retrieve project metadata dynamically via REST APIs.

Technical Implementation:

- REST integration with GitHub API for repository fetching.
- Spring Boot backend for project metadata storage.
- Frontend filtering logic using JavaScript.

Outcome:

The Projects Hub encourages collaborative software development and portfolio visibility, aligning with open-source learning practices.

3.9.14 Digital Resources Hub

Figure 3.14: Digital Resources Interface

The Digital Hub provides curated access to online learning materials like video lectures, e-books, datasets, and tools.

Process Description:

1. Students can explore YouTube-linked video lectures embedded directly within the LMS.
2. The search bar dynamically filters available courses or tools.
3. Backend APIs maintain categorized entries under digital_resources table.
4. Each lecture or course can be played inline, reducing the need for external navigation.

Technologies Used:

- YouTube API integration for embedded playback.
- AJAX-based live search and refresh.
- Dynamic iframe rendering for video lessons.

Outcome:

The Digital Hub extends the platform's functionality beyond traditional library use, offering self-learning and academic enrichment resources in one centralized interface.

3.9.15 Institutional Repository

Figure 3.15: Institutional Repository Module

The Institutional Repository is a centralized digital archive where academic materials such as project reports, theses, syllabi, and publications are stored.

Operational Workflow:

1. Authorized users upload files categorized by type, department, and year.
2. Records are stored in the institutional_repo table.
3. Admins manage uploads and validate the accuracy of metadata.
4. Users can filter items using dynamic dropdowns and perform searches across departments.

Technical Integration:

- File upload handled through MultipartFile in Spring Boot.
- Files stored on server directories with path mapping in the database.
- Secure access control ensures that only authenticated users can upload.

Outcome:

This repository supports institutional data archiving, ensuring preservation and accessibility of important academic content.

3.9.16 Newspaper Archives

Figure 3.16: Newspaper and Article Archive Module

The Newspaper Hub digitizes daily and historical newspapers to aid in research and general awareness.

Process Flow:

1. Articles are stored in a database with fields for title, publication, year, and type.
2. Users can search, sort, and filter based on publication or category (e.g., education, politics, technology).
3. The "Search Web" option integrates with online search APIs for real-time news updates.

Outcome:

This feature creates an academic news repository, helping students stay informed about current

events and trends relevant to their studies.

3.9.17 Trending AI Tools Portal

Figure 3.17: AI Tools Portal — 2025 Interface

The Trending AI Tools module is a unique addition to the platform, designed to introduce users to modern AI tools and productivity platforms.

Functional Overview:

- Displays curated tools categorized by function (Design, Writing, Audio, Video, Presentations, etc.).
- Includes "Student Mode" to recommend education-friendly tools.
- Each card links to external platforms like ChatGPT, Gamma.app, and RunwayML.
- Built with responsive grid layout for both dark and light themes.

Technical Description:

- Static dataset hosted on the LMS backend.
- JavaScript-based filtering and category selection.
- The interface ensures dynamic scalability for future tool additions.

Outcome:

This component integrates modern technological awareness into the LMS, promoting digital literacy and enabling students to explore AI-driven learning ecosystems.

3.9.18 Outcome Summary

The extended modules of the *E-Platform for Library Resources* demonstrate the versatility of full-stack development applied in an educational ecosystem.

The system evolved beyond a traditional Library Management System into a comprehensive institutional learning and resource management platform, integrating:

- Academic and research content (Institutional & Research Repositories)
- Student productivity features (Resume Builder, Projects Hub)
- Digital education support (Video Lectures, E-Resources)
- Real-time academic engagement (AI Tools and News Archives)

Overall Impact:

- Centralized academic infrastructure.
- Increased digital resource accessibility.
- Enhanced learning experience and institutional modernization.

3.10 Activity Repository and Deployment Link

The complete source code of the E-Platform for Library Resources Activity is maintained in a public GitHub repository for version control and demonstration purposes.

- GitHub Repository: https://github.com/venkat-0127/digital-library.git
- Live Deployment (Render): https://digital-library-4bne.onrender.com

These links allow access to the working web application and full Activity documentation, demonstrating successful backend integration and deployment on cloud infrastructure.

3.11 Skills Acquired

Technical Skills

- Advanced proficiency in Java, Spring Boot, and Maven for backend development.
- Frontend development experience using HTML, CSS, and JavaScript.
- Practical understanding of RESTful API design, development, and testing.
- Experience in database management with MySQL and Spring Data JPA.
- Exposure to API security, email automation, and payment gateway integration.

Soft Skills

- Enhanced communication through regular virtual meetings and code reviews.
- Strengthened teamwork and coordination during collaborative debugging sessions.
- Improved problem-solving and analytical thinking through error resolution.
- Learned professional ethics, punctuality, and deadline management.

3.12 Conclusion

The internship at Sandspace Technologies Pvt. Ltd. provided a comprehensive exposure to full-stack software development.

Through the design and development of the *E-Platform for Library Resources*, I gained practical knowledge of integrating multiple technologies into a unified, functional web system.

This internship helped me:

- Strengthen my core programming skills in Java and database management.
- Understand the lifecycle of a real-world web application.
- Develop a professional approach to teamwork, communication, and technical problem-solving.

Overall, this chapter covered the aim of internship, architecture of work, working principle, application, activity taken down application developed& used. The experience was invaluable in preparing me for a future career in software development, with a focus on Full Stack Java Engineering.

CHAPTER – 4

This chapter details the activity log book of the internship course carried out during internship and is presentenced in the form of 8 weeks.

4.1 ACTIVITY LOG FOR THE FIRST WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 19/05/25	Aim to know about the 3-2 summer internship and organizations who offering internships, which was conducted meeting by our college.	The learning outcomes from the discussion is gaining the knowledge about values of internship and organizations.	
Day – 2 20/05/25	Aim to select the organization and domain, in this free hand is given to us to select.	The learning outcomes from the Day -2 is to select the organization and domain.	
Day – 3 21/05/25	Aim to enroll to the organization and domain which was selected by us.	The learning outcomes from the Day -3 is to enrollment.	
Day – 4 22/05/25	Aim to onboarding us to the organization and more in detail about the internship and need to upload lecture notes in google form created by our college.	The learning outcomes from the meeting conducted by college is onboarding us and in detail about internship and google form to know our status.	
Day – 5 23/05/25	Aim to know about the organization and facilities offered by it for better internship experience.	The learning outcomes from the discussion is gaining the knowledge about organization and facilities given by them.	
Day – 6 24/05/25	Aim to know about the domain-java and where it is using and why it is trending.	The learning outcomes is to know about java and running applications in it and why it is trending.	
Day – 7 25/05/25	Aim to clear doubts and need to upload lecture notes in google form.	The learning outcomes is to clarification of doubts and upload notes.	

WEEKLY REPORT FOR WEEK - 1

WEEK – 1 (From Dt 19/05/2025 to Dt 25/05/2025)

Objective of the Activity Done:

Detailed Report:

At the beginning of the internship process, a college meeting was conducted to introduce the 3-2 summer internship program and provide details about the various organizations offering internship opportunities. This session helped in understanding the importance and value of internships and how they contribute to practical learning. The next step was to select an organization and domain of interest, allowing students to choose freely based on their career goals and technical preferences.

After finalizing the choices, the next task was to enrol in the selected organization and domain, officially beginning the internship journey. Following enrolment, an onboarding session was conducted by the organization, where the internship structure, rules, and expectations were explained in detail. The college also required uploading lecture notes and internship details in a Google Form to track student participation and progress.

In the following days, I learned more about the organization and the facilities provided to ensure a good internship experience. Later, the focus shifted to learning about the Java domain, understanding its uses, real-world applications, and why it continues to be a trending technology in software development.

Finally, a doubt clarification session was held, where all queries were resolved, and lecture notes were uploaded again in the Google Form as part of the regular internship documentation process.

4.2 ACTIVITY LOG FOR THE SECOND WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 26/05/25	Aim to learn about basics of java and what is java full stack & college meeting.	The learning outcomes of the session is to learn basic concepts of and about java full stack & college meeting also held.	
Day – 2 27/05/25	Aim to learn about variables, Data types & Non – primitive data types.	The learning outcomes of the session is to learn about variables, data types & Non – primitive data types.	
Day – 3 28/05/25	Aim to learn about Java tutorial from Geeksforgeeks it was given us task to complete it.	The learning outcomes of the session is assigned a work that is to complete the java tutorial from Geeksforgeeks.	
Day – 4 29/05/25	Aim to learn about java oops concept Abstraction & Encapsulation.	The learning outcomes of the session is to learn about oops concept Abstraction & Encapsulation.	
Day – 5 30/05/25	Aim to attend the online meeting from the college R&C division.	The learning outcomes of the session is track our internship status and clarifying our doubts.	
Day – 6 31/05/25	Aim to learn about remaining oops concept Inheritance & Polymorphism.	The learning outcomes of the session is to learn remaining oops concept.	
Day – 7 01/06/25	Aim to clear doubts and need to upload lecture notes in google form.	The learning outcomes is to clarification of doubts and upload notes.	

WEEKLY REPORT FOR WEEK - 2

WEEK – 2 (From Dt 26/05/22025 to Dt 01/06/2025)

Objective of the Activity Done:

Detailed Report:

At the beginning of the internship, the focus was on understanding the basics of Java and gaining a clear idea about what Java Full Stack Development means. The session also included a college meeting where the internship objectives and learning path were discussed. After that, I learned about variables, data types, and non-primitive data types, which helped me understand how data is stored and managed in Java programs.

In the next session, I was assigned a task to complete a Java tutorial from GeeksforGeeks, which helped strengthen my foundational knowledge through practical exercises and examples. Later, I learned about important Object-Oriented Programming (OOPs) concepts such as Abstraction and Encapsulation, understanding how they improve code structure, security, and reusability.

An online meeting was also attended, conducted by the college's R&C division, where we tracked internship progress and clarified our technical doubts. Following that, I continued learning the remaining OOPs concepts — Inheritance and Polymorphism, which provided a better understanding of how classes and objects interact in Java.

Finally, the week ended with a doubt-clearing session, and I uploaded the lecture notes through Google Forms, completing the documentation work for the week.

4.3 ACTIVITY LOG FOR THE THIRD WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 02/06/25	Aim to learn about HTML from basics to moderate.	The learning outcomes of the session is to learn about html from basics to moderate.	
Day –2 03/06/25	Aim to learn about CSS from basics to moderate.	The learning outcomes of the session is to learn about CSS from basics to moderate.	
Day – 3 04/06/25	Aim to learn about the Bootstrap framework which is used mainly for frontend design.	The leaning outcomes of the session is to learn about bootstrap which is useful for the frontend design.	
Day – 4 05/06/25	Aim to learn about the Java script from basics to moderate.	The learning outcomes of the session is to learn Java script in detail.	
Day – 5 06/06/25	Aim to attend the online meeting from the college R&C division & need to upload lecture notes.	The learning outcomes of the session is tracking our internship status and clarifying our doubts & upload lecture notes.	
Day – 6 07/06/25	Aim to learn about the Node.js & React.js from the basics to moderate.	The learning outcomes of the session is to learn node & react js in detail.	
Day – 7 08/06/25	Aim to learn about the frontend integration with learned from the above technologies.	The learning outcomes of the session is to explain the integration of frontend with learned technologies.	

WEEKLY REPORT FOR WEEK - 3

WEEK – 3 (From Dt 02/06/2025 to Dt 08/06/2025)

Objective of the Activity Done:

Detailed Report:

At the start of the internship, the focus was on learning HTML from basics to a moderate level, understanding how to structure web pages, use tags, and build simple layouts. This created a strong foundation for web development. The next step was to learn CSS, which helped in designing and styling web pages by adding colors, layouts, and visual effects to make the interface more attractive and user-friendly.

After gaining a good grasp of HTML and CSS, I moved on to learning the Bootstrap framework, which made it easier to design responsive and well-structured layouts using prebuilt classes and components. Following that, I started learning JavaScript, focusing on how to make web pages interactive and dynamic by manipulating elements and handling user events.

During the week, I also attended an online meeting conducted by the college's R&C division, where internship progress was discussed, doubts were clarified, and lecture notes were uploaded for documentation purposes. Later, I learned about Node.js and React.js, exploring how backend logic and frontend UI can work together efficiently.

Finally, I focused on frontend integration, combining all the learned technologies to understand how a complete and interactive web application is developed and connected across different layers.

4.4 ACTIVITY LOG FOR THE FORTH WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 09/06/25	Aim to learn about the spring initializr.io.	The learning outcomes of the session is to learn spring initializer, know how to create and install the project structure	
Day –2 10/06/25	Aim to learn about the Spring Boot framework for backend.	The learning outcomes of the session is to learn Spring boot framework for backend.	
Day – 3 11/06/25	Aim to learn about the overall integration of backend.	The learning outcomes of the session is to learn backend integration.	
Day – 4 12/06/25	Aim to learn about the what is data base	The learning outcomes of the session is to learn about the data bases.	
Day – 5 13/06/25	Aim to attend the online meeting from the college R&C division & need to upload lecture notes.	The learning outcomes of the session is tracking our internship status and clarifying our doubts & upload lecture notes.	
Day – 6 14/06/25	Aim to learn about the overall integration of full stack	The learning outcomes of the session is to learn about the integration of full stack.	
Day – 7 15/06/25	Aim to learn about the project structure.	The learning outcomes of the session is to learn in detail about the project structure.	

WEEKLY REPORT FOR WEEK - 4

WEEK – 4 (From Dt 09/06/2025 to Dt 15/06/2025)

Objective of the Activity Done:

Detailed Report:

At the beginning of the internship, the main focus was on learning Spring Initializr.io, understanding how to create and set up a new Spring Boot project structure. This provided a strong foundation for building backend applications efficiently. After that, the learning continued with an in-depth study of the Spring Boot framework, exploring how it simplifies backend development, handles dependencies, and provides an embedded server for easy execution.

Once familiar with the framework, the next step was to understand the overall backend integration process, including how different components like controllers, services, and repositories connect to make a complete system. Following this, I learned about databases, their importance in backend development, and how they store and manage application data effectively.

An online meeting was also attended, conducted by the college's R&C division, which helped in tracking internship progress, clarifying doubts, and uploading required lecture notes. Later, I learned about full-stack integration, gaining knowledge of how frontend and backend components communicate through APIs to build a complete web application.

Finally, I focused on understanding the project structure in detail, including folder organization, dependencies, and workflow, which helped prepare for further development stages.

4.5 ACTIVITY LOG FOR THE FIFTH WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 16/06/25	Aim to learn about the in detail about the project.	The learning outcomes of the session is to learn in detail about the project.	
Day –2 17/06/25	Aim to assigned to a project based on the internship & explained how output is required.	The learning outcomes of the session is assigned to a project University login portal & give requirement, how output is required.	
Day – 3 18/06/25	Aim to create a project development overview in a paper for gaining an idea about on the project.	The learning outcomes of the day is creating a paper-based approach to the project building.	
Day – 4 19/06/25	Aim to download the required tools & software's for the project development.	The learning outcome of the day is to download the required software's and tools for the assigned project.	
Day – 5 20/06/25	Aim to attend the online meeting conducted by the college R&C division & need to upload lecture notes.	The learning outcomes of the session is tracking our internship status and clarifying our doubts & upload lecture notes.	
Day – 6 21/06/25	Aim to start the work with the frontend based on the requirements and learned content.	The learning outcomes of the day is started the work in frontend with tools& technologies HTML, CSS, JS done work in visual studio.	
Day – 7 22/06/25	Aim to continue the development of frontend by solving the errors and debugging it.	The learning outcomes of the day continues the development of the frontend part of the project by debugging errors.	

WEEKLY REPORT FOR WEEK - 5

WEEK – 5 (From Dt 16/06/2025 to Dt 22/06/2025)

Objective of the Activity Done:

Detailed Report:

At the start of the internship, the focus was on understanding the project in detail — its objectives, workflow, and expected outcomes. This helped build a clear idea of what needed to be developed. Soon after, I was assigned to work on the University Login Portal project, where the requirements and expected output were clearly explained, giving a strong foundation to begin the development process.

Next, I created a project development overview on paper to visualize the structure and features before starting the actual implementation. This paper-based plan provided a clear roadmap for the upcoming development stages. Following that, I downloaded and set up all the necessary tools and software required for the project, including development environments and supporting libraries.

Later, I attended an online meeting conducted by the college's R&C division, which helped in tracking internship progress, clearing doubts, and ensuring proper documentation by uploading lecture notes. After completing the setup, I began working on the frontend using HTML, CSS, and JavaScript in Visual Studio Code, focusing on building the initial layouts and structure. Finally, I continued developing and refining the frontend by solving errors and debugging the code to ensure smooth functionality and a clean user interface.

4.6 ACTIVITY LOG FOR THE SIXTH WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 23/06/25	Aim to complete the all dashboards of frontend like index, student, admin, main.	The learning outcomes of the day is to complete the all dashboards in the project.	
Day –2 24/06/25	Aim to start the work of backend for the logic of login & student features.	The learning outcomes of the day is to complete the work of backend logics for login and student features.	
Day – 3 25/06/25	Aim to continue the work of backend in the logic of admin and faculty.	The learning outcomes of the day is to complete the work of backend logics and the debugging the errors.	
Day – 4 26/06/25	Aim to integration of frontend backend with using tools like postman.	The learning outcomes of the day is to complete integration of frontend and backend with tool postman and fixing the errors.	
Day – 5 27/06/25	Aim to attend the online meeting conducted by the college R&C division.	The learning outcomes of the session is concluding our internship and start the work of report and ppt for reviews.	
Day – 6 28/06/25	Aim to integrate the database to the project and debugging the errors.	The learning outcomes of the day is to integrate the MySQL database to the project & debug the errors.	
Day – 7 29/06/25	Aim to start the project in local host.	The learning outcomes of the day is to start the project in the local host and debugging all the issues to working condition.	

WEEKLY REPORT FOR WEEK - 6

WEEK – 6 (From Dt 23/06/2025 to Dt 29/06/2025)

Objective of the Activity Done:

Detailed Report:

At the beginning of this phase, the main objective was to complete the development of all frontend dashboards, including the index, student, admin, and main pages. Each page was carefully designed and structured to ensure a consistent user interface and smooth navigation throughout the application. The focus was on achieving a clean layout, responsive design, and role-based functionality for different users within the system.

Once the frontend was completed, the next step involved starting the backend development, particularly focusing on implementing the logic for login and student features. This included designing APIs for authentication, handling user data securely, and connecting the frontend forms with backend services. After completing the student-side functionality, work progressed toward the admin and faculty modules. The backend logic for these sections was implemented to manage tasks such as user handling, data access, and system control. During this process, several errors were encountered and debugged to ensure a stable backend structure.

Following the backend development, the integration of the frontend and backend was carried out using tools like Postman. The APIs were tested thoroughly to verify that the data flow between both ends worked seamlessly. Any integration errors or mismatches were identified and resolved to ensure smooth communication between the client and server. Later, an online meeting was attended, conducted by the college's R&C division, which served as a formal session to conclude the internship phase and prepare for the final project review. The meeting provided valuable insights into documentation standards and presentation expectations.

Subsequently, the database integration process was carried out by connecting the project to the MySQL database. Tables were created, relationships were established, and data retrieval and storage functionalities were tested. Debugging was performed to eliminate database-related issues and ensure proper synchronization with the backend. Finally, the project was successfully hosted and executed on localhost. This involved running the complete system locally, testing all modules in real-time, and resolving any remaining bugs to achieve a fully functional working environment.

Throughout this period, I learned how to structure a full-stack web application from the frontend to backend and database integration. It enhanced my skills in debugging, API testing, and deployment preparation, while also improving my ability to manage project workflow efficiently from start to finish.

4.7 ACTIVITY LOG FOR THE SEVENTH WEEK

Day& Date	Brief description of the daily activity	Learning Outcome	Person In-Charge Signature
Day – 1 30/06/25	Aim to sort out the all issues in the project overall integration part not completed at 29/06/25.	The learning outcomes of the day is to complete overall integration which is not completed.	
Day –2 31/06/25	Aim to modify the issues and updates which was said by the knowledge persons.	The learning outcomes of the day is to clean layout or structure the updates what need to update.	
Day – 3 01/07/25	Aim to solve the each and every update which was we structured to update.	The learning outcomes of the day is to update the one by one what we need to add in the project	
Day – 4 02/07/25	Aim to resolve the issues what was we get while doing the updates.	The learning outcomes of the day is to resolve the update issues in code.	
Day – 5 03/07/25	Aim to push the code which was developed till now in the GitHub.	The learning outcome of the day is to push the overall code in the GitHub.	
Day – 6 04/07/25	Aim to deploy the project in the open-source platforms like Render, Netlify	The learning outcome of the day is to deploy the code in the Render (deployment)	
Day – 7 05/07/25	Aim to change the Api's local to global to working code deployment.	The learning outcomes of the day is to change the Api's locally to globally.	

WEEKLY REPORT

WEEK – 7 (From Dt 30/06/25 to Dt 05/07/25)

Objective of the Activity Done: Detailed Report:

During the initial stage of the project, the main focus was on identifying and resolving all pending issues in the overall integration. The modules were not fully connected, so efforts were made to ensure smooth communication between the frontend and backend. This involved checking the data flow, fixing compatibility problems, and verifying that all project components worked together seamlessly. Once the integration was stable, the next step was to implement the suggestions and modifications provided by mentors and knowledgeable individuals. Their feedback helped refine the structure, improve the layout, and enhance the overall appearance and functionality of the project.

After restructuring, the next phase involved systematically solving and implementing each update that had been planned. Each enhancement was added step by step to ensure it aligned with the project's functionality and design goals. While making these updates, several issues and bugs appeared in the code, which were resolved through debugging and testing. The debugging process helped identify logic errors and improved the overall stability of the system.

Once all updates were successfully implemented and tested, the developed code was pushed to GitHub. Proper version control practices were followed to maintain a clean, organized, and collaborative code repository. After ensuring the latest version was uploaded, the focus shifted toward deploying the project on open-source platforms such as Render and Netlify. The deployment process involved connecting the repository, configuring the environment, and verifying that the project was live and functional online.

Finally, after successful deployment, the API connections were updated from local to global to ensure the application could function seamlessly on the hosted environment. This change enabled smooth communication between the frontend and backend servers in a live setting. Throughout the process, I learned important concepts such as project integration, code optimization, version control, deployment, and API configuration for real-world applications.

4.8 ACTIVITY LOG FOR THE EIGHTH WEEK

Day& Date	**Brief description of the daily activity**	**Learning Outcome**	**Person In-Charge Signature**
Day – 1	Aim to modify overall code to clean for the deployment.	The learning outcomes of the day is to clean package of code for deployment.	
Day –2	Aim to Attend to reviews which was conducted in college and preparation of documentation.	The learning outcomes of the day is to portray our work which was done in internship and preparation of reports.	
Day – 3	Aim to Attend to reviews which was conducted in college and preparation of documentation.	The learning outcomes of the day is to portray our work which was done in internship and preparation of reports.	
Day – 4	Aim to Attend to reviews which was conducted in college and preparation of documentation.	The learning outcomes of the day is to portray our work which was done in internship and preparation of reports.	
Day – 5	Aim to Attend to reviews which was conducted in college and preparation of documentation.	The learning outcomes of the day is to portray our work which was done in internship and preparation of reports.	
Day – 6	Aim to Attend to reviews which was conducted in college and preparation of documentation.	The learning outcomes of the day is to portray our work which was done in internship and preparation of reports.	
Day – 7	Aim to Attend to reviews which was conducted in college and preparation of documentation.	The learning outcomes of the day is to portray our work which was done in internship and preparation of reports.	

WEEKLY REPORT
WEEK – 8

Objective of the Activity

Done: Detailed Report:

In the final phase of the project, the main focus was on modifying and cleaning the overall code in preparation for deployment. All unnecessary files, redundant functions, and unused dependencies were removed to ensure a clean and optimized package. This helped make the codebase lighter, more organized, and ready for hosting on deployment platforms. Proper testing was also done to verify that all modules were functioning correctly after optimization.

After completing the code cleanup, the following days were mainly dedicated to attending the internal reviews conducted by the college and preparing the project documentation. During the review sessions, the project progress and features were presented, and feedback was received from the evaluators. This helped in understanding how to professionally explain the technical aspects, demonstrate the working of the system, and highlight key outcomes achieved during the internship.

The documentation work included compiling the complete report, covering objectives, methodologies, implementation details, challenges faced, and final outcomes. This process continued over several days to ensure the report was well-structured, accurate, and aligned with academic requirements. Through these activities, I learned how to effectively present and document technical work, communicate project progress clearly, and maintain proper records of internship activities for evaluation and submission.

This chapter cover the daily details of the internship work completio

CHAPTER 5: OUTCOMES DESCRIPTION

5.1 Describe the work environment you have experienced

During my internship, I experienced a highly professional and supportive work environment that played a key role in enhancing my learning process. The people interactions were very encouraging, as mentors and coordinators were approachable, willing to clarify doubts, and motivated us to perform well. There was clear communication regarding roles and responsibilities, which helped me understand my tasks without confusion. The organization maintained a healthy balance of guidance and independence, allowing me to explore and learn while still following protocols and procedures.

The facilities provided were adequate and well-maintained, which created a smooth working experience. Access to necessary software tools, resources, and documentation ensured that I could carry out my tasks efficiently. The workspace was clean, well-ventilated, and organized, providing a comfortable environment to stay focused for long hours. Technical support was also available, ensuring that errors or issues during project execution could be resolved without major delays.

Time management and discipline were highly emphasized, as daily and weekly schedules were clearly defined. This taught me the importance of working within deadlines, maintaining consistency, and balancing tasks systematically. The organization followed proper protocols for communication, project documentation, and reporting, which gave me a structured approach to work.

Another important aspect was teamwork and mutual support. Even though individual tasks were assigned, there was a spirit of collaboration among peers, where knowledge-sharing and helping one another were encouraged. This not only strengthened professional relationships but also created a harmonious environment where learning became enjoyable. Socialization and interaction during meetings allowed us to build rapport with mentors and other interns, promoting healthy professional networking.

Overall, the work environment was highly motivating and learner-friendly, combining discipline with mutual support. It provided me with valuable exposure to professional practices, clear role clarity, and efficient processes. The harmonious relationships, team spirit, and structured protocols contributed to a positive and productive experience, ensuring that the internship was both professionally enriching and personally satisfying.

5.2 Describe the real time technical skills you have acquired

During the internship, I worked on a E-Platform for Library resources, which focused on authentication, user role management, and secure access to university resources. This project helped me gain in-depth experience in both backend and frontend development, along with database integration.

Job-Related Technical Skills

- **Frontend Development**: Built login and dashboard interfaces using HTML, CSS, and JavaScript, ensuring a user-friendly experience.
- **Backend Development:** Used Java and Spring Boot to implement secure login logic and role-based features for students, faculty, and admin users.
- **Database Skills:** Worked with MySQL to handle user records, authentication data, and role-based access storage.
- **Project Design and Planning:** Created a paper-based development overview to plan workflows and structure before implementation.
- **Debugging and Integration:** Gained experience in debugging backend logic, frontend issues, and integrating the database with the backend system.

Hands-On Experience

- Configured Spring Initializr.io and project dependencies to set up the development environment.
- Downloaded and used essential tools and software to build and run the project.
- Integrated the project and successfully deployed it on localhost, testing all modules for functionality.

Continuous Learning

- Learned how to analyse requirements and convert them into working features.
- Improved understanding of protocols, project structures, and best practices in software development.

Conclusion:

This project gave me real-time exposure to authentication systems, backend logic, and role-based access control. It not only enhanced my technical knowledge but also improved my problem-solving, debugging, and documentation skills essential for professional software development.

5.3 Describe the managerial skills you have acquired

During my internship, I not only developed technical expertise but also acquired several important managerial skills that strengthened my overall professional competency. These skills helped me manage tasks effectively, work collaboratively, and deliver results within deadlines.

Planning and Goal Setting:

I learned to plan my weekly tasks systematically by breaking down project requirements into smaller, achievable goals. This approach improved my ability to prioritize activities and track progress, ensuring steady advancement towards project completion.

Leadership and Teamwork:

Through collaborative discussions with peers and mentors, I gained experience in both leading and contributing as a team member. I learned how to share responsibilities, delegate tasks, and support others, which enhanced the overall efficiency of group work.

Behaviour and Workmanship:

The internship emphasized professional behaviour, discipline, and quality of work. I developed a sense of responsibility and accountability, ensuring that my contributions met organizational standards.

Decision Making and Problem-Solving:

I enhanced my decision-making abilities by analysing problems, evaluating different approaches, and implementing the most effective solutions. This was particularly useful in debugging technical issues and selecting tools for integration.

Performance Analysis and Improvement:

Regular feedback sessions and college review meetings allowed me to analyse my performance weekly. This helped me identify strengths, address weaknesses, and improve competencies step by step.

Overall, the internship provided me with strong managerial skills such as planning, leadership, teamwork, time management, decision-making, and continuous performance improvement. These skills are equally valuable as technical knowledge, as they prepare me to take responsibility, work collaboratively, and perform effectively in professional environments.

5.4 Describe how you could improve your communication skills

During my internship, I realized that communication plays a vital role in both technical and managerial activities. To further improve my communication skills, I plan to focus on several key areas that will strengthen my ability to interact effectively in professional environments.

Oral and Written Communication:

I aim to enhance my oral communication by practicing clear pronunciation, tone, and fluency during meetings and presentations. For written communication, I will improve the way I structure emails, reports, and documentation to make them more precise, professional, and easy to understand.

Conversational Abilities and Confidence:

To become more confident in conversations, I will practice active participation in discussions, express my thoughts clearly, and learn to manage anxiety during public speaking. Improving my extempore speaking skills will help me respond quickly and effectively in unplanned situations.

Listening and Understanding:

Good communication also involves attentive listening. I plan to strengthen my ability to understand others by maintaining eye contact, avoiding interruptions, and asking clarifying questions when needed. This will ensure that I am understood correctly and can also understand others better.

Articulation and Conversation Management:

I will practice summarizing key points in discussions, ensuring that the main ideas are conveyed clearly. Additionally, I will work on closing conversations politely and maintaining professional niceties such as greeting, thanking, and appreciating others.

Overall:

By improving oral and written communication, building confidence, managing anxiety, enhancing listening skills, and practicing professional etiquette, I can develop strong communication skills. These improvements will help me collaborate better, present my ideas effectively, and maintain positive professional relationships in future career opportunities.

5.5 Describe how could you could enhance your abilities in group discussions, participation in teams, contribution as a team member, leading a team/activity.

Enhancing Abilities in Group Discussions and Team Participation

Group discussions and teamwork are essential parts of professional growth, and I believe there is always scope for improvement in these areas. To enhance my abilities, I plan to focus on developing active participation, constructive contribution, and effective leadership skills.

Group Discussions:

I aim to strengthen my performance in group discussions by improving my ability to listen carefully, analyse different viewpoints, and respond with relevant, concise points. I will practice maintaining clarity of thought, staying confident, and contributing meaningfully without dominating or being passive. This balance will help me engage productively in discussions.

Participation in Teams:

To become an effective participant, I will actively contribute ideas, share knowledge, and take responsibility for assigned tasks. At the same time, I will remain open to feedback and respect the opinions of others, ensuring that the team functions harmoniously toward common goals.

Contribution as a Team Member:

I plan to focus on collaboration by helping peers when required, encouraging them, and ensuring that my work supports the team's success. By improving my coordination, adaptability, and willingness to take initiative, I can add greater value to team projects.

Leading a Team/Activity:

To enhance my leadership abilities, I will practice planning activities, setting clear goals, and motivating team members to perform their best. I will also work on decision-making, conflict resolution, and time management skills to ensure smooth execution of tasks. Leading with empathy and fairness will help me earn trust and build strong professional relationships.

Overall:

By improving active participation in discussions, contributing responsibly as a team member, and practicing fair and motivating leadership, I can significantly enhance my abilities in teamwork and group activities. These improvements will make me more confident, collaborative, and effective in both academic and professional environments.

5.6 Describe the technological developments you have observed and relevant to thesubject area of training

While working on the E-Platform for library resources, I observed several technological advancements in the area of secure authentication systems and digital access management. These technologies are widely used to manage academic portals and online services for institutions.

- **Spring Boot for Authentication**: Modern login systems use Spring Security within Spring Boot to handle authentication, session management, and role-based access, ensuring secure and scalable backends.
- **Database Integration:** The use of MySQL for storing user credentials and roles highlights how relational databases are vital for maintaining secure and structured login systems.
- **Encryption Techniques:** I observed how password encryption and hashing are essential developments to protect sensitive login data against cyber threats.
- **Frontend Technologies:** Interfaces designed using HTML, CSS, and JavaScript with frameworks like Bootstrap provide responsive and accessible login pages across devices.
- **API and Middleware:** REST APIs enable smooth communication between frontend login forms and backend authentication services, which is a standard practice in digital portals today.
- **Deployment Trends:** University portals are increasingly being deployed on cloud servers, ensuring high availability, remote access, and scalability for large numbers of users.
- **Digital Transformation in Education:** The adoption of single sign-on (SSO), integration with student information systems, and OTP/email verification reflect the latest advancements in digital university login systems.

Conclusion:

These technological developments demonstrate how university login portals are evolving with secure authentication frameworks, cloud deployment, and responsive UI designs, ensuring safe, reliable, and user-friendly access to academic resources.

Student Self Evaluation of the ShortTerm Internship

Student Name: Thota. Venkata Rao **Registration No:** 22NA1A103

Term of Internship: 6 weeks **From:** 20/05/2025 **To:** 06/07/2025

Date of Evaluation:

Organization Name & Address: Sandspace Technologies Private Limited
Address: 2nd Floor, S B I, NCK Plaza, NTR Circle, Vijayawada, Andhra Pradesh 520010

Please rate your performance in the following areas:

Rating Scale: **Letter grade of CGPA calculation to be provided**

1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	**OVERALL PERFORMANCE**	**1**	**2**	**3**	**4**	**5**

Date: **Signature of the Student**

Evaluation by the Supervisor of the Intern Organization

Student Name: Thota. Venkata Rao **Registration No:** 22NA1A6103

Term of Internship: 8 weeks **From:** 20/05/2025 **To:** 06/07/2025

Date of Evaluation:

Organization Name & Address: Sandspace Technologies Private Limited
Address **:** 2nd Floor, S B I, NCK Plaza, NTR Circle, Vijayawada, Andhra Pradesh 520010
Name & Address of the Supervisor
with Mobile Number:

Please rate the student's performance in the following areas:

Please note that your evaluation shall be done independent of the Student's self-evaluation

Rating Scale: 1 is lowest and 5 is highest rank

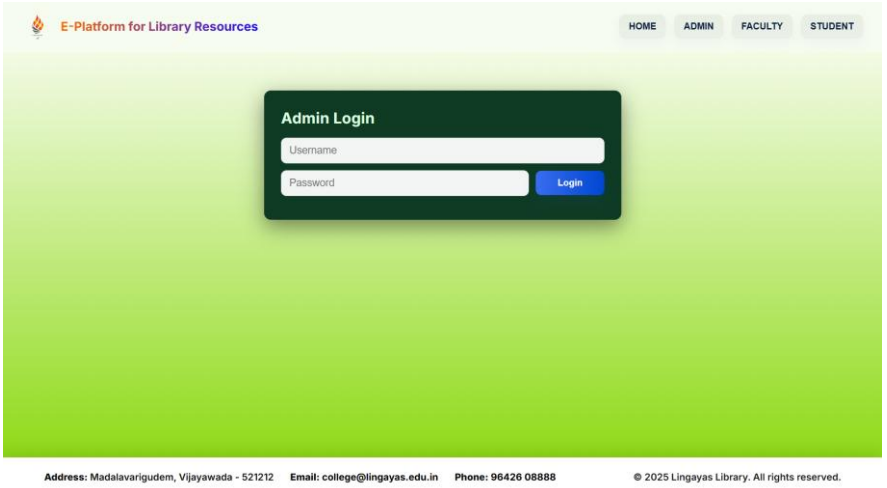
1	Oral communication	1	2	3	4	5
2	Written communication	1	2	3	4	5
3	Proactiveness	1	2	3	4	5
4	Interaction ability with community	1	2	3	4	5
5	Positive Attitude	1	2	3	4	5
6	Self-confidence	1	2	3	4	5
7	Ability to learn	1	2	3	4	5
8	Work Plan and organization	1	2	3	4	5
9	Professionalism	1	2	3	4	5
10	Creativity	1	2	3	4	5
11	Quality of work done	1	2	3	4	5
12	Time Management	1	2	3	4	5
13	Understanding the Community	1	2	3	4	5
14	Achievement of Desired Outcomes	1	2	3	4	5
15	**OVERALL PERFORMANCE**	**1**	**2**	**3**	**4**	**5**

Date: **Signature of the Supervisor**

Photos And Videos

Fig.3.1 Main homepage of the E-Platform. Users can choose roles Admin, Faculty, & Student and view library info.

Fig.3.2 Admin enters username and password to access the dashboard for managing books and users.



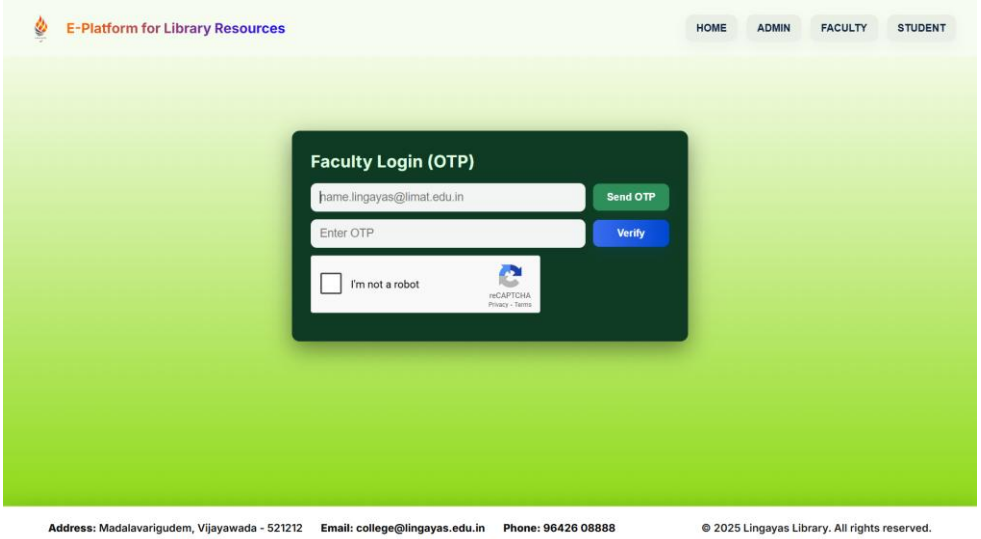


Fig.3.3 Faculty login using college email. OTP is sent for secure access to library

Fig.3.4 Students log in using institutional email with OTP verification to access the dashboard.

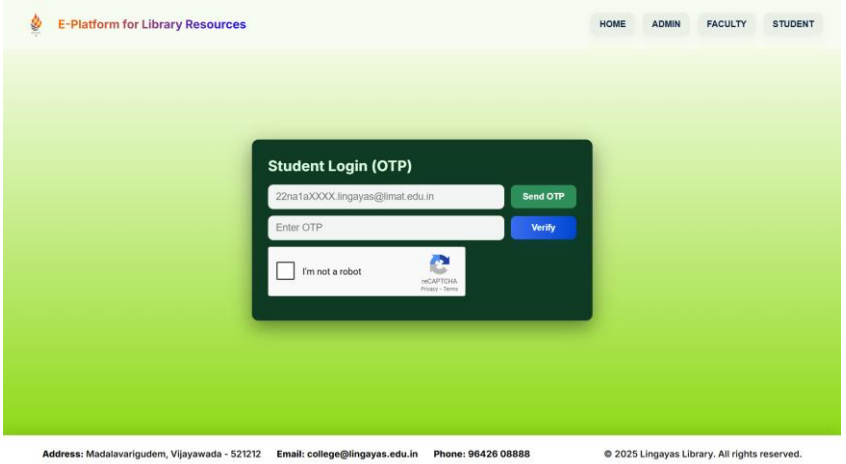
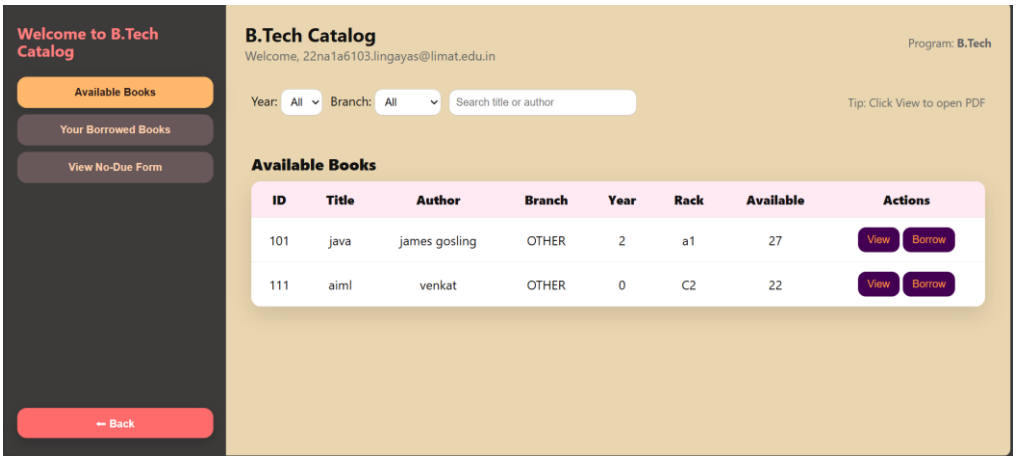


Fig.3.5 Main dashboard showing options like B.Tech, M.Tech/MBA, Research, Career, and Digital Resources.

Fig.3.6 This figure shows the responsive version of the dashboard optimized for mobile and tablet

Fig.3.7 Lists all available books with filters. Users can view or borrow books



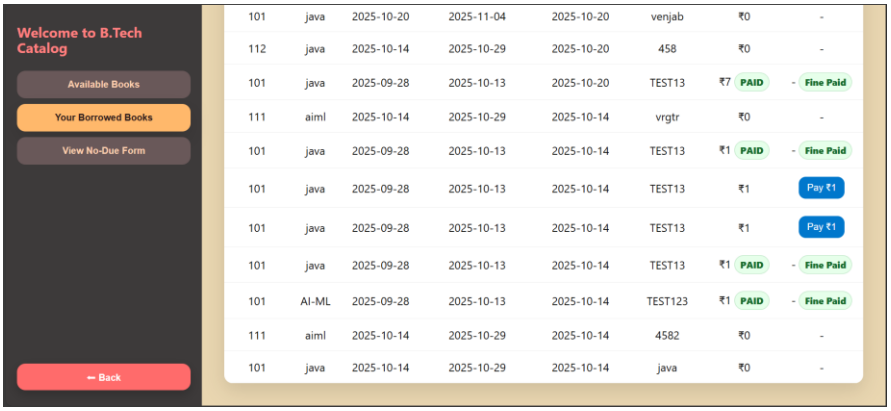
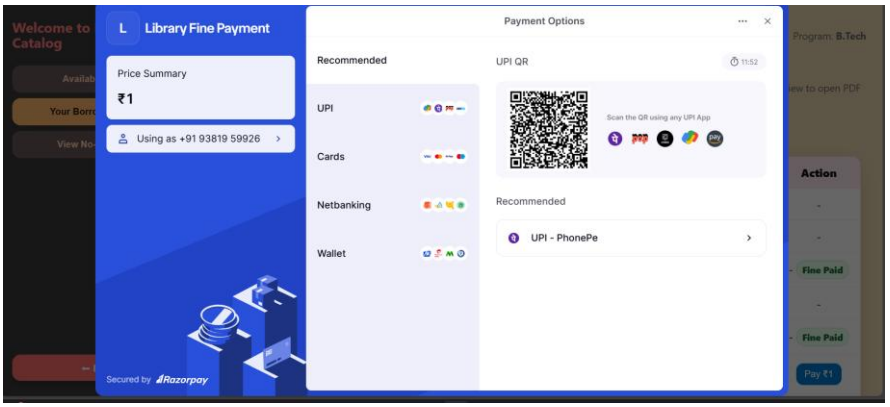


Fig.3.8 Shows user's borrowed books, return dates, and fine details.

Fig.3.9 Online fine payment using Razorpay UPI or card for student transactions.



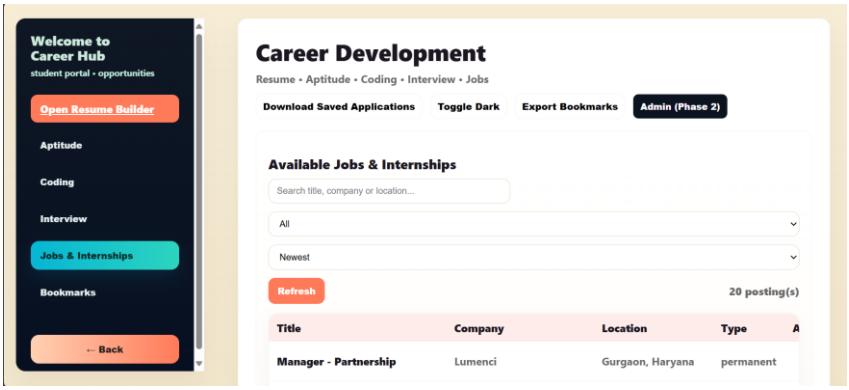
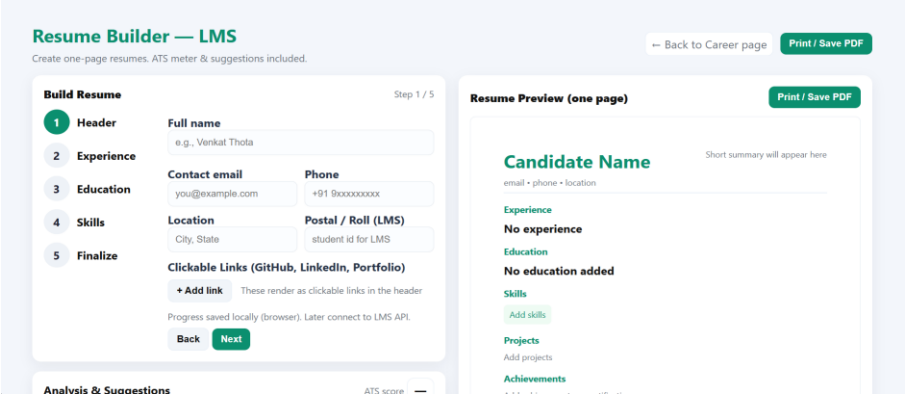


Fig.3.10 Career module showing job listings, internships, and resume builder access.

Fig.3.11 Tool to create and download a one-page professional resume.



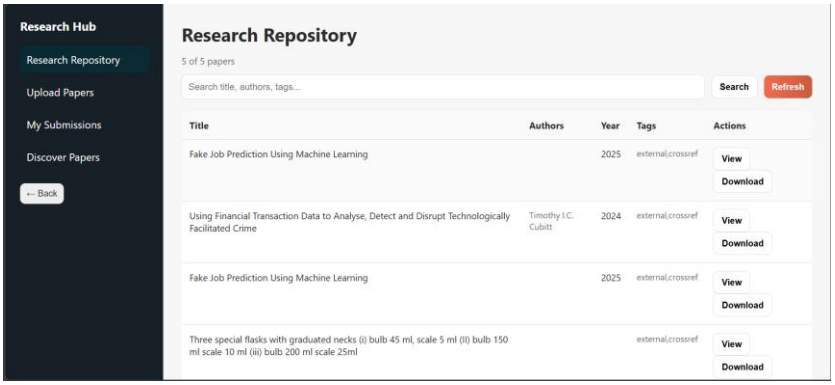
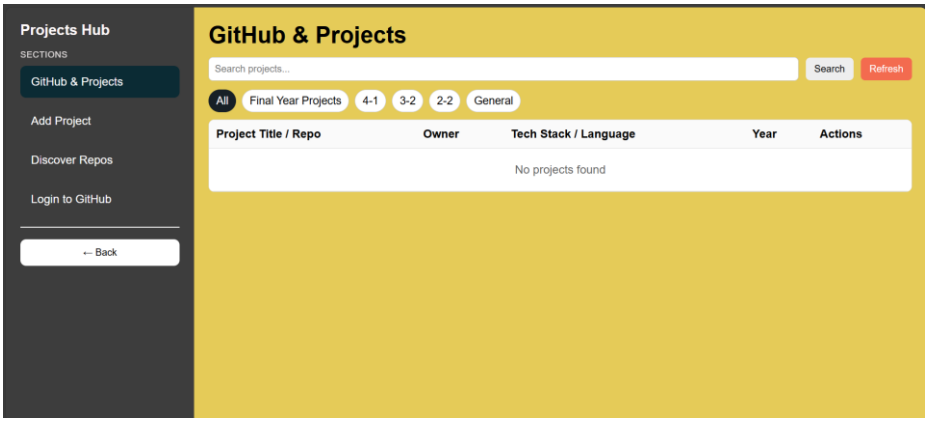


Fig.3.12 Repository of research papers for viewing and downloading academic work.

Fig.3.13 Lists and connects student GitHub projects by semester and technology.



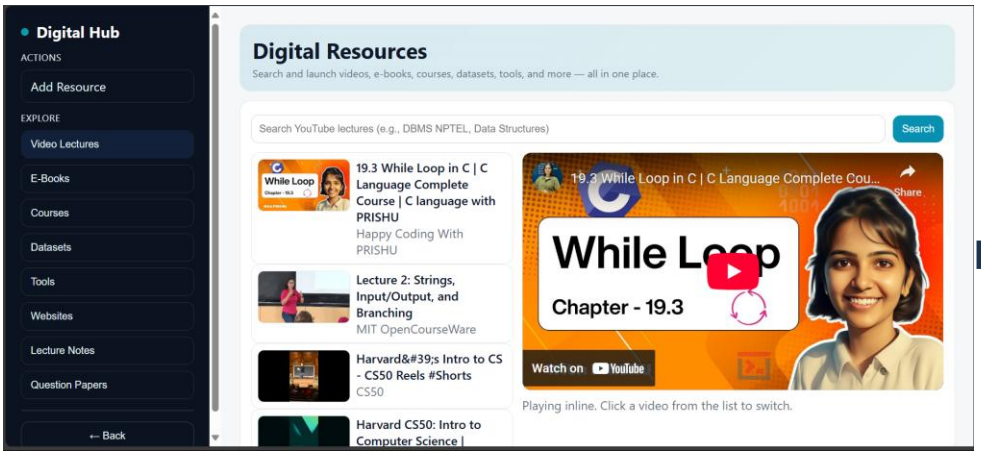
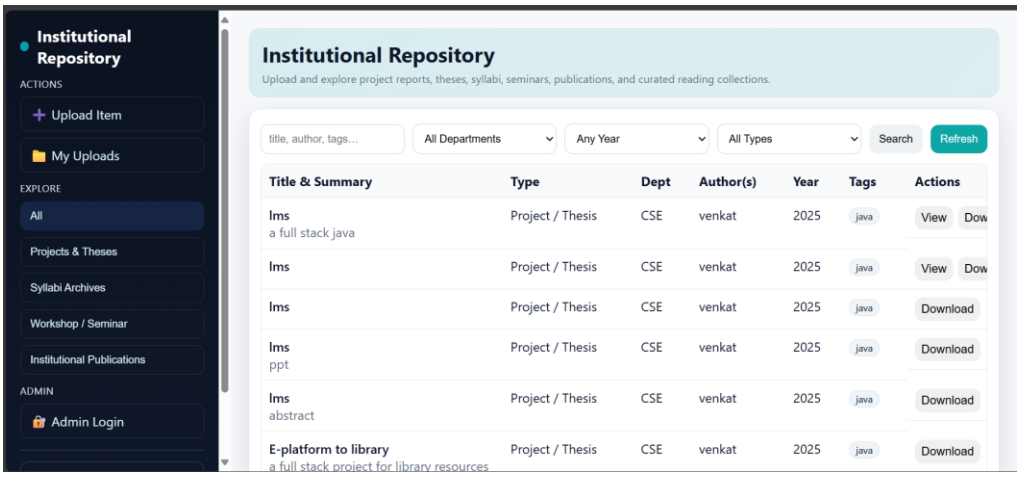


Fig.3.14 Contains e-books, videos, and online course materials for learning.

Fig.3.15 Stores project reports, theses, and seminar documents for departments.



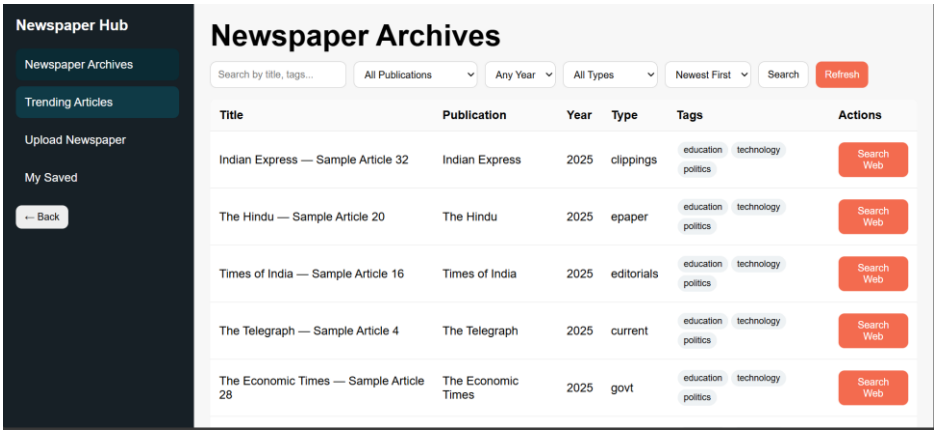


Fig.3.16 Shows academic and current news articles for reading and research.

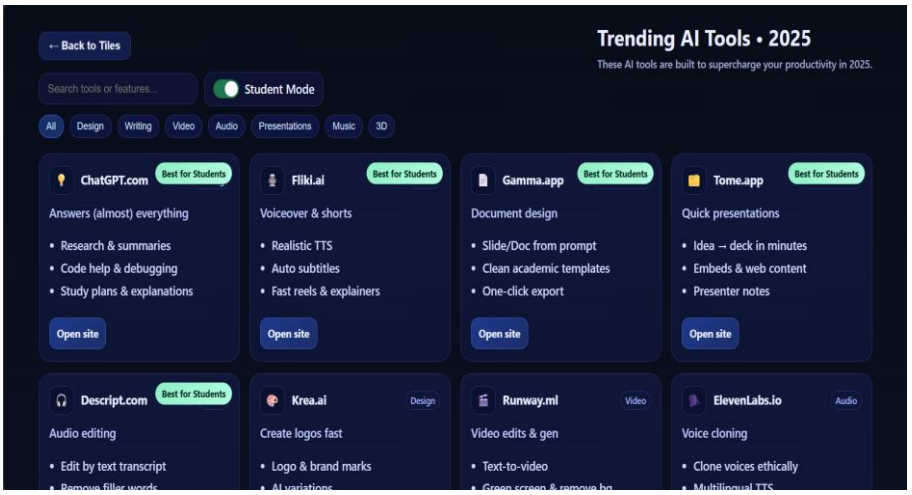


Fig.3.17 Displays trending AI tools and apps useful for students and researchers.

EVALUATION

Internal Evaluation for Short Term Internship(Virtual)

Objectives:

- To integrate theory and practice.
- To learn to appreciate work and its function towards the future.
- To develop work habits and attitudes necessary for job success.
- To develop communication, interpersonal and other critical skills in thefuture job.
- To acquire additional skills required for the world of work.

Assessment Model:

- There shall only be internal evaluation.
- The Faculty Guide assigned is in-charge of the learning activities of the students and for the comprehensive and continuous assessment of the students.
- The assessment is to be conducted for 100 marks.
- The number of credits assigned is 4. Later the marks shall be converted into grades and grade points to include finally in the SGPA and CGPA.
- The weightings shall be:
    - Activity Log 25 marks
    - Internship Evaluation 50marks
    - Oral Presentation 25 marks
- Activity Log is the record of the day-to-day activities. The Activity Log is assessed on an individual basis, thus allowing for individual members withingroups to be assessed this way. The assessment will take into consideration the individual student's involvement in the assigned work.
- While evaluating the student's Activity Log, the following shall be considered –

  a. The individual student's effort and commitment.

  b. The originality and quality of the work produced by the individualstudent.

  c. The student's integration and co-operation with the work assigned.

  d. The completeness of the Activity Log.
- The Internship Evaluation shall include the following components and basedon Weekly Reports and Outcomes Description

  a. Description of the Work Environment.

b. Real Time Technical Skills acquired.
c. Managerial Skills acquired.
d. Improvement of Communication Skills.
e. Team Dynamics
f. Technological Developments recorded.

INTERNAL ASSESSMENT STATEMENT

Name Of the Student: Thota Venkata Rao

Programme of Study: Bachelor of Technology

Year of Study: 4th YEAR

Group: AI&ML

Register No/H.T. No: 22NA1A6103

Name of the College: Lingayas Institute of Management & Technology

University: Jawaharlal Nehru Technological University, Kakinada

Sl.No	***Evaluation Criterion***	***Maximum Marks***	***Marks Awarded***
1.	Activity Log	25	
2.	Internship Evaluation	50	
3.	Oral Presentation	25	
	GRAND TOTAL	100	

Date: **Signature of the Faculty Guide**

Certified by

Date: **Signature of the Head of the Department/Principal**

Seal:

ANDHRA PRADESH
STATE COUNCIL OF HIGHER EDUCATION

(A Statuory Body of the Government of Andhra Pradesh)

2nd, 3rd, 4th and 5th floors, Neeladri Towers, Sri Ram Nagar, 6th Battalion Road
Atmakur (V)Mangalagiri (M), Guntur, Andhra Pradesh, Pin - 522 503
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